



THE INFLUENCE OF DIFFERENTIATION STRATEGY, MARKET-DRIVEN AND ARTIFICIAL INTELLIGENCE MARKETING ON SUSTAINABLE COMPETITIVE ADVANTAGE WITH SCHOOL IMAGE AS AN INTERVENING VARIABLE MODERATE BY VOCATIONAL SCHOOL CULTURE IN THE CITY OF BALIKPAPAN

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ABSTRACT

Objective: This research aims to investigate the impact of differentiation strategies, market-based approaches, and artificial intelligence marketing on sustainable competitive advantage, with school image as an intervening variable moderated by school culture at Balikpapan Private Vocational Schools.

Theoretical Framework: Differentiation Strategy and Sustainable Competitive Advantage; Market Driven and Sustainable Competitive Advantage; Artificial Intelligence Marketing and Sustainable Competitive Advantage; School Image as an Intervening Variable; School Culture as a Moderating Variable By using these theoretical approaches, this research aims to explore how differentiation strategies, market orientation, and artificial intelligence-based marketing can influence sustainable competitive advantage in private vocational schools in Balikpapan City, with school image as an intervention variable and school culture as a moderating variable.

Method: This research was designed as research using a quantitative descriptive approach. The data analysis technique uses SEM (Structural Equation Modeling) using AMOS software. This research can also be said to be research with an inductive paradigm, namely testing from a sample which will then be generalized to the parent population. The population in this study were 12th-grade students at Balikpapan Private Vocational School. Researchers chose class 12 because students at that class level can be said to have lived for quite a long time and experienced the learning process at school since entering (class 10) and have made adjustments until they reach the final level (class 12). So, grade 12 students are students who already have self-confidence towards various components of school, be it towards teachers, friends, or the applicable rules. The population and samples were taken through school clusters with the criteria for schools that have been accredited B.

Results and Discussion: The results of multivariate outlier detection based on Table 5.14 show that fourteen observations have a mahalonobis d-squared value greater than the table chi-square limit of 58.30, namely respondents number 4, 5, 8, 9, 10, 12, 13, 14, 17, 18, 21, 22, 23, and 28 so that these fourteen observations (respondents) were indicated as outliers, and were excluded from the analysis. Next, the data will be reanalyzed without including outliers, so the sample size will be 223 (237-14=223). This size of 223 still meets the sample size criteria explained in the previous subchapter. this research confirms the importance of differentiation, market orientation and artificial intelligence-based marketing strategies in building sustainable competitive advantages for private vocational schools in Balikpapan City, with school image as a key factor that mediates this influence. An integrated and responsive approach to market needs and the application of innovative technology will increase the school's competitiveness and reputation sustainably.

Research Implications: Practical Implications: Differentiation Strategy Development: 1. Schools: Educational institutions can implement effective differentiation strategies to increase sustainable competitiveness. This includes offering unique programs, innovative teaching methods, and extracurricular activities that are different from other schools 2. Education Policy: Policymakers can encourage schools to develop their special strengths, by providing support in the form of training and resources.

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Improved School Image: 1. School Management: School managers must focus on efforts to strengthen the school's image, such as improving facilities, achieving high academic achievement, and building good relationships with the local community. 2. Marketing Communications: School marketing teams should leverage this positive image in their communications strategies to attract more potential students. Market-Based Approach: 1. Curriculum Strategy: Schools must continually adapt to market and industry needs, ensuring that their curriculum is relevant to today's work demands. 2. Collaboration with Industry: Partnerships with local companies and industry can help adapt educational programs to the real needs of the world of work. Application of Artificial Intelligence Based Marketing: 1. Technology Implementation: Schools should adopt AI technology in their marketing efforts to increase personalization and efficiency of communication with prospective students and parents. 2. Training and Development: Specialized training is required for marketing staff to optimize the use of AI tools.

Theoretical Implications: Development of Educational Marketing Theory: This research enriches the literature on marketing strategies in vocational education, especially regarding how differentiation and market-based approaches influence sustainable competitive advantage. The Role of School Image as a Mediating Variable: The results of this study confirm the importance of school image as a significant mediating variable between differentiation strategies, market-based approaches, AI marketing, and sustainable competitive advantage. This adds to the theoretical understanding of influence mechanisms in educational contexts. School Culture as Moderator: This study also highlights the important role of school culture in moderating the relationship between AI marketing and sustainable competitive advantage, providing new insights into how cultural factors influence the effectiveness of technology marketing strategies. Affected Areas or Sectors: 1. Vocational Education Sector: Provides practical guidance for vocational schools to increase competitiveness through technology-based differentiation and marketing strategies. 2. Education Policy Development: Offers recommendations for policymakers to support differentiation and innovation initiatives in the curriculum. 3. School Management: Provides implementation strategies that can assist school management in improving the image and attractiveness of their schools. 4. Educational Marketing and Technology: Informing educational marketing practitioners about the benefits and challenges of using AI in their marketing strategies. This research, therefore, makes a significant practical and theoretical contribution, supporting the development of more effective and innovative educational policies and practices.

Originality/Value: This research contributes to the literature by highlighting originality through an innovative approach in combining differentiation strategies, market-based approaches, and artificial intelligence marketing to enhance sustainable competitive advantage in the vocational education sector. The relevance and value of this research are evidenced by how the results can have a significant impact on school management practices, strengthen the image of institutions, and guide educational policies to be more responsive to market and technological needs..

Keywords: Differentiation Strategy, Market Driven, Artificial Intelligence Marketing, School Image, Sustainable Competitive Advantage.

A INFLUÊNCIA DA ESTRATÉGIA DE DIFERENCIAÇÃO, DO MARKETING DE MERCADO E DE INTELIGÊNCIA ARTIFICIAL NA VANTAGEM COMPETITIVA SUSTENTÁVEL COM A IMAGEM ESCOLAR COMO VARIÁVEL INTERVENIENTE MODERADA PELA CULTURA ESCOLAR PROFISSIONAL NA CIDADE DE BALIKPAPAN

RESUMO

Objetivo: Esta pesquisa tem como objetivo investigar o impacto de estratégias de diferenciação, abordagens baseadas no mercado e marketing de inteligência artificial em vantagem competitiva sustentável, com imagem escolar como uma variável intermediária moderada pela cultura escolar nas Escolas Profissionais Privadas de Balikpapan.

Estrutura Teórica: Estratégia de Diferenciação e Vantagem Competitiva Sustentável; Vantagem Competitiva Orientada para o Mercado e Sustentável; Marketing de Inteligência Artificial e Vantagem Competitiva Sustentável; Imagem Escolar como Variável Interveniante; Cultura Escolar como Variável Moderadora. Ao usar estas abordagens teóricas, esta pesquisa visa explorar como estratégias de diferenciação, orientação de mercado e marketing baseado em inteligência artificial podem influenciar a vantagem competitiva sustentável em escolas profissionais privadas na cidade de Balikpapan, com a imagem escolar como uma variável de intervenção e a cultura escolar como uma variável moderadora.



Método: Esta pesquisa foi projetada como pesquisa usando uma abordagem descritiva quantitativa. A técnica de análise de dados usa SEM (Structural Equation Modeling) usando o software AMOS. Esta pesquisa também pode ser considerada como uma pesquisa com um paradigma indutivo, ou seja, testando a partir de uma amostra que será então generalizada para a população-mãe. A população deste estudo era de estudantes do 12º ano da Escola Profissional Privada de Balikpapan. Os pesquisadores escolheram a classe 12 porque se pode dizer que os alunos dessa classe viveram por um bom tempo e experimentaram o processo de aprendizagem na escola desde que entraram (classe 10) e fizeram ajustes até chegarem ao nível final (classe 12). Assim, alunos da 12ª série são estudantes que já têm autoconfiança em vários componentes da escola, seja em relação a professores, amigos, ou as regras aplicáveis. A população e as amostras foram coletadas por meio de agrupamentos escolares com os critérios para escolas credenciadas B.

Resultados e Discussão: Os resultados da detecção de outliers multivariados com base na Tabela 5.14 mostram que quatorze observações têm um valor d-quadrado de mahalanobis maior que o limite qui-quadrado da tabela de 58.30, ou seja, respondentes número 4, 5, 8, 9, 10, 12, 13, 14, 17, 18, 21, 22, 23 e 28, de modo que essas quatorze observações (respondentes) foram indicadas como outliers, e foram excluídas da análise. Em seguida, os dados serão reanalisados sem incluir outliers, portanto o tamanho da amostra será 223 ($237-14=223$). Este tamanho de 223 ainda atende aos critérios de tamanho da amostra explicados no subcapítulo anterior. Esta pesquisa confirma a importância da diferenciação, orientação de mercado e estratégias de marketing baseadas em inteligência artificial na construção de vantagens competitivas sustentáveis para escolas profissionais privadas na cidade de Balikpapan, com a imagem escolar como um fator chave que medeia essa influência. Uma abordagem integrada e responsiva às necessidades do mercado e a aplicação de tecnologia inovadora aumentarão a competitividade e a reputação da escola de forma sustentável.

Implicações da pesquisa: Implicações práticas: Estratégia de diferenciação Desenvolvimento: 1. Escolas: As instituições educacionais podem implementar estratégias de diferenciação eficazes para aumentar a competitividade sustentável. Isso inclui a oferta de programas únicos, métodos de ensino inovadores e atividades extracurriculares diferentes de outras escolas. 2. Política de educação: os legisladores podem incentivar as escolas a desenvolverem os seus pontos fortes especiais, prestando apoio sob a forma de formação e recursos. Imagem Escolar Melhorada: 1. Gestão Escolar: Os gestores escolares devem se concentrar nos esforços para fortalecer a imagem da escola, como melhorar as instalações, alcançar alto desempenho acadêmico e construir bons relacionamentos com a comunidade local. 2. Comunicações de Marketing: As equipes de marketing da escola devem alavancar esta imagem positiva em suas estratégias de comunicação para atrair mais estudantes potenciais. Abordagem Baseada No Mercado: 1. Estratégia Curricular: As escolas devem se adaptar continuamente às necessidades do mercado e da indústria, garantindo que seu currículo seja relevante para as demandas de trabalho de hoje. 2. Colaboração com a indústria: Parcerias com empresas locais e a indústria podem ajudar a adaptar programas educacionais às necessidades reais do mundo do trabalho. Aplicação de Inteligência Artificial Baseada Marketing: 1. Implementação de tecnologia: as escolas devem adotar a tecnologia de IA em seus esforços de marketing para aumentar a personalização e a eficiência da comunicação com futuros alunos e pais. 2. Formação e desenvolvimento: é necessária formação especializada para o pessoal de marketing otimizar a utilização das ferramentas de IA.

Implicações Teóricas: Desenvolvimento da Teoria de Marketing Educacional: Esta pesquisa enriquece a literatura sobre estratégias de marketing no ensino profissional, especialmente no que diz respeito a como a diferenciação e abordagens baseadas no mercado influenciam a vantagem competitiva sustentável. O Papel da Imagem Escolar como Variável de Mediação: Os resultados deste estudo confirmam a importância da imagem escolar como uma variável mediadora significativa entre estratégias de diferenciação, abordagens baseadas no mercado, marketing de IA e vantagem competitiva sustentável. Isso contribui para a compreensão teórica dos mecanismos de influência em contextos educacionais. Cultura Escolar como Moderadora: Este estudo também destaca o importante papel da cultura escolar na moderação da relação entre marketing de IA e vantagem competitiva sustentável, fornecendo novas percepções sobre como fatores culturais influenciam a eficácia das estratégias de marketing de tecnologia. Áreas ou setores afetados: 1. Setor do ensino profissional: fornece orientações práticas para as escolas profissionais aumentarem a competitividade através de estratégias de diferenciação e comercialização baseadas na tecnologia. 2. Desenvolvimento de políticas de educação: oferece recomendações para os formuladores de políticas para apoiar iniciativas de diferenciação e inovação no currículo. 3. Gestão Escolar: Fornece estratégias de implementação que podem ajudar a gestão escolar na melhoria da imagem e atratividade de suas escolas. 4. Marketing e Tecnologia Educacional: Informar os profissionais de marketing educacional sobre os benefícios e desafios do uso da IA em suas estratégias de marketing. Esta pesquisa, portanto, dá uma contribuição prática e teórica significativa, apoiando o desenvolvimento de políticas e práticas educacionais mais eficazes e inovadoras.



Originalidade/valor: Esta pesquisa contribui para a literatura destacando a originalidade através de uma abordagem inovadora na combinação de estratégias de diferenciação, abordagens baseadas no mercado e marketing de inteligência artificial para aumentar a vantagem competitiva sustentável no setor de educação profissional. A relevância e o valor desta pesquisa são evidenciados por como os resultados podem ter um impacto significativo nas práticas de gestão escolar, fortalecer a imagem das instituições e orientar as políticas educacionais para serem mais sensíveis às necessidades tecnológicas e de mercado.

Palavras-chave: Estratégia de Diferenciação, Mercado, Marketing de Inteligência Artificial, Imagem Escolar, Vantagem Competitiva Sustentável.

LA INFLUENCIA DE LA ESTRATEGIA DE DIFERENCIACIÓN, EL MARKETING IMPULSADO POR EL MERCADO Y LA INTELIGENCIA ARTIFICIAL SOBRE LA VENTAJA COMPETITIVA SOSTENIBLE CON LA IMAGEN ESCOLAR COMO VARIABLE INTERVENTORA MODERADA POR LA CULTURA ESCOLAR VOCACIONAL EN LA CIUDAD DE BALIKPAPAN

RESUMEN

Objetivo: Esta investigación tiene como objetivo investigar el impacto de las estrategias de diferenciación, los enfoques basados en el mercado y el marketing de inteligencia artificial en la ventaja competitiva sostenible, con la imagen de la escuela como una variable interviniente moderada por la cultura escolar en las Escuelas Vocacionales Privadas de Balikpapan.

Marco teórico: Estrategia de diferenciación y ventaja competitiva sostenible; Ventaja competitiva sostenible y impulsada por el mercado; Marketing de inteligencia artificial y ventaja competitiva sostenible; Imagen escolar como variable de intervención; Cultura escolar como variable moderadora. Mediante el uso de estos enfoques teóricos, esta investigación tiene como objetivo explorar cómo las estrategias de diferenciación, la orientación al mercado y el marketing basado en la inteligencia artificial pueden influir en la ventaja competitiva sostenible en las escuelas vocacionales privadas en la ciudad de Balikpapan, con la imagen de la escuela como una variable de intervención y la cultura escolar como una variable moderadora.

Método: Esta investigación fue diseñada como investigación utilizando un enfoque cuantitativo descriptivo. La técnica de análisis de datos utiliza SEM (Modelado de Ecuaciones Estructurales) utilizando el software AMOS. Esta investigación también puede decirse que es una investigación con un paradigma inductivo, a saber, la prueba de una muestra que luego se generalizará a la población de padres. La población en este estudio eran estudiantes de 12° grado en la Escuela Vocacional Privada de Balikpapan. Los investigadores eligieron la clase 12 porque se puede decir que los estudiantes en ese nivel de clase han vivido durante bastante tiempo y han experimentado el proceso de aprendizaje en la escuela desde que ingresaron (clase 10) y han hecho ajustes hasta que alcanzaron el nivel final (clase 12). Por lo tanto, los estudiantes de grado 12 son estudiantes que ya tienen confianza en sí mismos hacia varios componentes de la escuela, ya sea hacia los maestros, amigos o las reglas aplicables. La población y las muestras se tomaron a través de grupos escolares con los criterios para las escuelas que han sido acreditadas B.

Resultados y discusión: Los resultados de la detección de valores atípicos multivariados basados en la Tabla 5.14 muestran que catorce observaciones tienen un valor de mahalonobis al cuadrado d mayor que el límite de chi cuadrado de la tabla de 58.30, es decir, los encuestados número 4, 5, 8, 9, 10, 12, 13, 14, 17, 18, 21, 22, 23 y 28, de modo que estas catorce observaciones (encuestados) se indicaron como valores atípicos y se excluyeron del análisis. A continuación, los datos se volverán a analizar sin incluir valores atípicos, por lo que el tamaño de la muestra será de 223 ($237-14 = 223$). Este tamaño de 223 aún cumple con los criterios de tamaño de muestra explicados en el subcapítulo anterior. esta investigación confirma la importancia de la diferenciación, la orientación al mercado y las estrategias de marketing basadas en la inteligencia artificial en la construcción de ventajas competitivas sostenibles para las escuelas vocacionales privadas en la ciudad de Balikpapan, con la imagen escolar como un factor clave que media esta influencia. Un enfoque integrado y sensible a las necesidades del mercado y la aplicación de tecnologías innovadoras aumentarán la competitividad y la reputación de la escuela de forma sostenible.

Implicaciones de la investigación: Implicaciones prácticas: Desarrollo de estrategias de diferenciación: 1. Escuelas: Las instituciones educativas pueden implementar estrategias de diferenciación efectivas para aumentar la competitividad sostenible. Esto incluye ofrecer programas únicos, métodos de enseñanza innovadores y actividades extracurriculares que son diferentes de otras escuelas. 2. Política educativa: Los responsables políticos pueden alentar a las escuelas a desarrollar sus puntos fuertes especiales, proporcionando apoyo en forma de capacitación y recursos. Mejora de la imagen de la escuela: 1. Administración de la escuela: Los gerentes de la



escuela deben centrarse en los esfuerzos para fortalecer la imagen de la escuela, como mejorar las instalaciones, lograr un alto rendimiento académico y construir buenas relaciones con la comunidad local. 2. Comunicaciones de marketing: Los equipos de marketing escolar deben aprovechar esta imagen positiva en sus estrategias de comunicación para atraer a más estudiantes potenciales. Enfoque Basado En El Mercado: 1. Estrategia curricular: Las escuelas deben adaptarse continuamente a las necesidades del mercado y de la industria, asegurándose de que su plan de estudios sea relevante para las demandas de trabajo de hoy. 2. Colaboración con la industria: Las asociaciones con las empresas locales y la industria pueden ayudar a adaptar los programas educativos a las necesidades reales del mundo del trabajo. Aplicación de Inteligencia Artificial Basado en Marketing: 1. Implementación de tecnología: Las escuelas deben adoptar la tecnología de IA en sus esfuerzos de marketing para aumentar la personalización y la eficiencia de la comunicación con los futuros estudiantes y padres. 2. Formación y desarrollo: Se requiere formación especializada para que el personal de marketing optimice el uso de las herramientas de IA.

Implicaciones Teóricas: Desarrollo de la Teoría del Marketing Educativo: Esta investigación enriquece la literatura sobre estrategias de marketing en la educación vocacional, especialmente con respecto a cómo la diferenciación y los enfoques basados en el mercado influyen en la ventaja competitiva sostenible.

El papel de la imagen escolar como variable mediadora: Los resultados de este estudio confirman la importancia de la imagen escolar como variable mediadora significativa entre las estrategias de diferenciación, los enfoques basados en el mercado, el marketing de IA y la ventaja competitiva sostenible. Esto se suma a la comprensión teórica de los mecanismos de influencia en contextos educativos. La cultura escolar como moderadora: Este estudio también destaca el importante papel de la cultura escolar en la moderación de la relación entre el marketing de IA y la ventaja competitiva sostenible, proporcionando nuevos conocimientos sobre cómo los factores culturales influyen en la efectividad de las estrategias de marketing tecnológico. Zonas o sectores afectados: 1. Sector de la educación profesional: proporciona orientación práctica para que las escuelas de formación profesional aumenten la competitividad a través de estrategias de diferenciación y comercialización basadas en la tecnología. 2. Desarrollo de políticas educativas: Ofrece recomendaciones para que los responsables de las políticas apoyen las iniciativas de diferenciación e innovación en el plan de estudios. 3. Gestión escolar: Proporciona estrategias de implementación que pueden ayudar a la gestión escolar a mejorar la imagen y el atractivo de sus escuelas. 4. Marketing y Tecnología Educativa: Informar a los profesionales del marketing educativo sobre los beneficios y desafíos de usar la IA en sus estrategias de marketing. Esta investigación, por lo tanto, hace una contribución práctica y teórica significativa, apoyando el desarrollo de políticas y prácticas educativas más efectivas e innovadoras.

Originalidad/Valor: Esta investigación contribuye a la literatura al resaltar la originalidad a través de un enfoque innovador al combinar estrategias de diferenciación, enfoques basados en el mercado y marketing de inteligencia artificial para mejorar la ventaja competitiva sostenible en el sector de la educación vocacional. La relevancia y el valor de esta investigación se evidencian en cómo los resultados pueden tener un impacto significativo en las prácticas de gestión escolar, fortalecer la imagen de las instituciones y orientar las políticas educativas para que sean más sensibles a las necesidades del mercado y tecnológicas.

Palabras Clave: Estrategia de Diferenciación, Impulsada por el Mercado, Inteligencia Artificial, Marketing, Imagen Escolar, Ventaja Competitiva Sostenible.

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1 INTRODUCTION

In the context of the digital era and increasingly fierce global competition, educational institutions, including Vocational High Schools (SMK), are faced with major challenges to maintain their competitive advantage and relevance in the market. This background underlies the urgency of this research.

In Balikpapan City, as in many other places, vocational schools play a vital role in preparing young people with relevant skills to enter the competitive job market. However, with rapid changes in industrial and technological demands, as well as shifts in consumer preferences, SMKs are faced with pressure to continuously innovate their strategies. In this case, the concept of sustainable competitive advantage becomes very important. Competitive advantage is achieved when an organization can create greater value for customers compared to competitor (Viswanathan Kumar & Pansari, 2016). Differentiation strategy, which involves creating unique, value-added products or services, is one of the main approaches to achieving sustainable competitive advantage (Kaleka & Morgan, 2017).

This research summarizes several important elements in the context of vocational school competitive advantage in Balikpapan City. First, is strategic differentiation, namely the ability to offer something unique and different from competitors. Second, a strong market orientation places market needs and desires as the main focus of strategy development (Racela, 2014). Third, the use of artificial intelligence in marketing efforts provides the opportunity to optimize interactions with prospective students and other stakeholders. Artificial intelligence in marketing can increase efficiency and effectiveness through deeper data analysis, better personalization, and automation of marketing processes. AI helps in understanding consumer behavior better and delivering more targeted messages, which can ultimately strengthen an organization's competitive position (Huang & Rust, 2021). Market orientation is an approach that emphasizes the importance of understanding customer needs and wants and integrating this information into organizational strategy and operations. Their research shows that organizations that have a strong market orientation tend to perform better in the long term (Ramayah et al., 2011).

However, in addition to these factors, it is important to consider the role of school image as a representation of an institution's perception and reputation, which can influence student acceptance and support from the community. Furthermore, school culture also has a significant role in shaping strategy and policy implementation, as well as in driving overall organizational success. Organizational culture is a pattern of basic assumptions discovered, developed, or



created by a group of people to learn to overcome problems of external adaptation and internal integration (Büschgens et al., 2013). An organizational culture that supports innovation and adaptation can facilitate the implementation of new strategies and technologies, which in turn can increase competitive advantage (Alsabbagh & Khalil, 2017). Institutional image is the perception held by the public about an organization (Elsbach, 2003). A positive image can increase loyalty and support from various stakeholders, including prospective students and the wider community (Glavas & Godwin, 2013). Research shows that a strong institutional image can be a valuable asset that strengthens an organization's competitiveness (Bankins & Waterhouse, 2019).

By combining these elements, this research aims to investigate how strategic differentiation, market orientation, and artificial intelligence-based marketing can influence sustainable competitive advantage for vocational schools in Balikpapan City. In this context, school image is raised as an intervention variable that strengthens or weakens the relationship between these factors, while school culture is considered a moderating factor that can strengthen or weaken their influence. It is hoped that the combination of these factors will provide in-depth and applicable insight for vocational school managers to improve the competitiveness and reputation of their institutions (Chintalapati & Pandey, 2021).

2 LITERATURE REVIEW

2.1 DIFFERENTIATION STRATEGY AND SUSTAINABLE COMPETITIVE ADVANTAGE

Differentiation strategy is one of the main strategies proposed by (Samarrokhi et al., 2014) in his theory of competitive advantage. This strategy involves creating a product or service that is unique and different from competitors to attract consumers. In an educational context, differentiation can take the form of a unique curriculum, superior educational facilities, or innovative extra-curricular programs. (Mitroulis & Kitsios, 2017) asserts that valuable, rare, inimitable, and non-substitutable resources are the key to achieving sustainable competitive advantage, and differentiation strategies help create these resources. Private Vocational Schools in Balikpapan City can utilize this strategy to offer a unique educational experience, attract more students, and maintain a competitive edge.



2.2 MARKET DRIVEN AND SUSTAINABLE COMPETITIVE ADVANTAGE

Market orientation or market-driven is an approach that places market needs and desires at the centre of organizational strategy. (Vikas Kumar et al., 2011) and (Mujanah et al., 2022) show that organizations with a strong market orientation tend to perform better because they are more responsive to changes in the market and consumer needs. In the vocational school context, market orientation can be translated as a deep understanding of the needs of students and parents as well as adapting educational programs to suit the demands of the job market. Thus, private vocational schools in Balikpapan City that adopt this approach can increase their competitive advantage.

2.3 ARTIFICIAL INTELLIGENCE MARKETING AND SUSTAINABLE COMPETITIVE ADVANTAGE

Artificial intelligence (AI)--based marketing has become an important tool in increasing marketing efficiency and effectiveness. According to (Hossain et al., 2022), AI in marketing enables in-depth data analysis, personalization, and automation of marketing processes. However, their research also suggests that AI's direct impact on competitive advantage may require further exploration (Ekasari et al., 2023). In education, AI can help vocational schools personalize communications with prospective students and parents, as well as improve the school's image through smart and targeted marketing strategies.

2.4 SCHOOL IMAGE AS AN INTERVENING VARIABLE

School image is the public's perception of the quality and reputation of a school. (Desiani et al., 2023) states that a positive institutional image can increase loyalty and support from various stakeholders. Recent research by (Riyadi & Arif, 2023) confirms that a strong image can be the main driver in creating a competitive advantage. In the context of private vocational schools in Balikpapan, effective AI-based differentiation, market orientation and marketing strategies can strengthen the school's image, which then contributes to sustainable competitive advantage.



2.5 SCHOOL CULTURE AS A MODERATING VARIABLE

School culture plays an important role in implementing strategies and achieving organizational goals. (Wu et al., 2019) show that an organizational culture that supports innovation and adaptation can increase the effectiveness of differentiation and market orientation strategies. A positive and innovative school culture can moderate the relationship between school image and competitive advantage, strengthening the positive impact of implemented strategies.

By using these theoretical approaches, this research aims to explore how differentiation strategies, market orientation, and artificial intelligence-based marketing can influence sustainable competitive advantage in private vocational schools in Balikpapan City, with school image as an intervention variable and school culture as a moderating variable.

3 RESULT

The results of multivariate outlier detection based on Table 5.14 show that fourteen observations have a mahalanobis d-squared value greater than the table chi-square limit of 58.30, namely respondents number 4, 5, 8, 9, 10, 12, 13, 14, 17, 18, 21, 22, 23, and 28 so that these fourteen observations (respondents) were indicated as outliers, and were excluded from the analysis. Next, the data will be reanalyzed without including outliers, so the sample size will be 223 ($237-14=223$). This size of 223 still meets the sample size criteria explained in the previous subchapter.

Table 1

Multivariate Outlier Test

Observation number	Mahalanobis d-squared
23	121.887
4	102.750
21	97.634
12	92.088
13	84.135
22	76.982
9	74.464
8	73.421
5	72.721
10	71.441
17	68.119
28	62.386

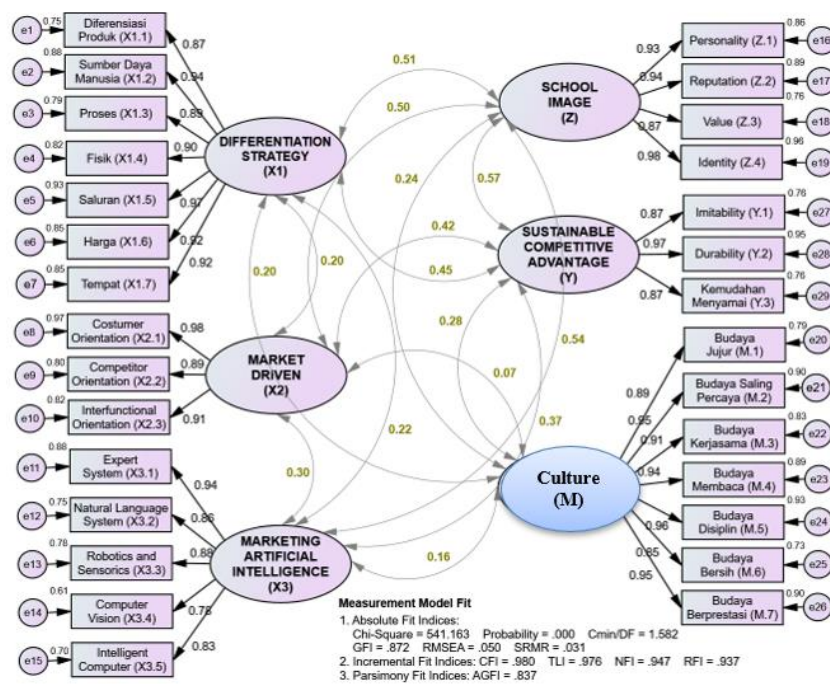


Observation number	Mahalanobis d-squared
18	61.205
14	61.008
2	57.695
7	55.783
26	53.585
19	53.041
⋮	
⋮	
119	25.858
226	25.804

The measurement model analysis was carried out simultaneously on all constructs, the estimation results of which are presented in Figure 1.

Figure 1

Measurement Model Estimation Results



Based on Figure 1, interpretation will then be carried out regarding the suitability of the measurement model, construct validity testing, and construct reliability testing.



Table 2

Fit Measure on Measurement Models

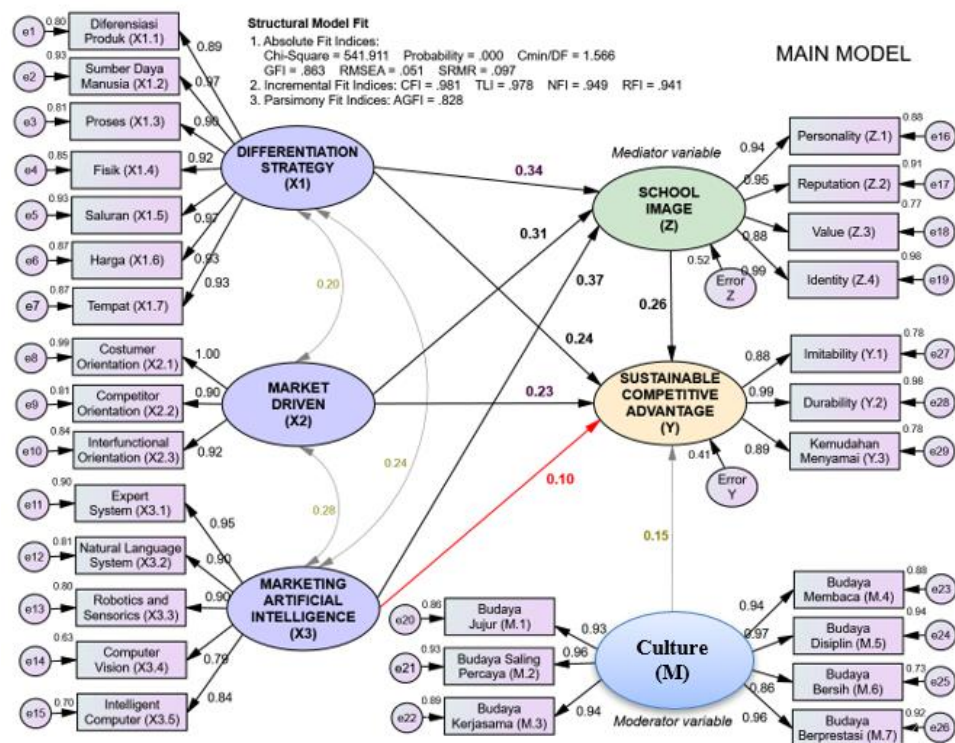
Fit Measure		Critical Value	Measurement model	
			Index value	Information
Absolute Fit Indices	Prob. χ^2 ^(a)	> 0,05	0,000	fit Even a good
	Cmin/df	≤ 3,00	1,582	Good fit
	GFI	≥ 0,90	0,872	Marginal fit
	RMSEA	≤ 0,08	0,050	Good fit
	SRMR	≤ 0,08	0,031	Good fit
Incremental Fit Indices	CFI	≥ 0,95	0,980	Good fit
	TLI	≥ 0,94	0,976	Good fit
	NFI	≥ 0,90	0,947	Good fit
	RFI	≥ 0,90	0,937	Good fit
Parsimony Fit Indices	AGFI	≥ 0,90	0,837	Marginal fit

^(a) In a model with a sample size of $n > 250$ or several indicators more than 25 ($m > 25$), the model is still fit even though the probability value is below 0.05 or even a good fit. (Hair *et al.*, 2018).

Table 2 shows that the results of the suitability test on the measurement model have produced appropriate criteria, both absolute fit indices, incremental fit indices, and parsimony fit indices. Thus, the measurement model can be accepted because the suitability of the model is good (good fit and marginal fit).

Figure 2

Main Model SEM Estimation Results





Tabel 3

Fit Measure on Structural Model

Fit Measure		Critical Value	Structural Model	
			Index value	Information
Absolute Fit Indices	Prob. χ^2 ^(a)	> 0,05	0,000	Even a good fit
	Cmin/DF	≤ 3,00	1,566	Good fit
	GFI	≥ 0,90	0,863	Marginal fit
	RMSEA	≤ 0,08	0,051	Good fit
	SRMR	≤ 0,08	0,087	Marginal fit
Incremental Fit Indices	CFI	≥ 0,95	0,981	Good fit
	TLI	≥ 0,94	0,978	Good fit
	NFI	≥ 0,90	0,949	Good fit
	RFI	≥ 0,90	0,941	Good fit
Parsimony Fit Indices	AGFI	≥ 0,90	0,828	Marginal fit
^(a) In a model with a sample size of n>250 or several indicators more than 25 (m>25), the model is still fit even though the probability value is below 0.05 or even a good fit. (Hair et al., 2018).				

Table 4 shows that the RZ2 value is 0.522, meaning that the percentage of influence of differentiation strategies, market-driven, and artificial intelligence marketing on school image at private vocational schools in Balikpapan City is 52.2 per cent, while 47.8 per cent is influenced by other variables. Furthermore, the RY2 value is 0.407, meaning that the percentage of influence of differentiation strategy, market driving, artificial intelligence marketing, and school image on the sustainable competitive advantage at private vocational schools in Balikpapan City is 40.7 per cent, while 59.3 per cent is influenced by other variables.

It is known that the total coefficient of determination (R2 total) is 0.468, this shows that the model developed in this research can explain around 46.8 per cent of the variability in the data. In another sense, the model in this research has quite good predictive relevance or is relevant to be used to predict sustainable competitive advantage in private vocational schools in Balikpapan City through differentiation strategies, market-driven, artificial intelligence marketing, and school image.



Tabel 4

Coefficient of Determination (R^2)

Influence Between Variables	R^2
$X1, X2, X3 \rightarrow Z$	$R_Z^2 = 0,522$
$X1, X2, X3, Z \rightarrow Y$	$R_Y^2 = 0,407$
$R^2 \text{ total} = 1 - \sqrt{[(1-R_Z^2) \times (1-R_Y^2)]}$ $= 1 - \sqrt{[(1-0,522) \times (1-0,407)]}$ $= 1 - 0,532 = \mathbf{0,468}$	
<u>Information:</u> X1: Differentiation Strategy Z: School Image X2: Market Driven M: Culture X3: Artificial intelligence marketing Y: Sustainable Competitive Advantage	

Tabel 5

Test Structural Relationships Between Variables

No	Direct path of influence	Std. Estimate	S.E.	C.R.	$P^{(a)}$	Hypothesis
1	$X1 \rightarrow Y$	0,238	0,055	3,764	0,005	H ₁ Accepted
2	$X1 \rightarrow Z$	0,341	0,051	6,529	0,021	H ₂ Accepted
3	$X2 \rightarrow Y$	0,231	0,053	3,849	0,004	H ₄ Accepted
4	$X2 \rightarrow Z$	0,311	0,049	6,286	0,013	H ₅ Accepted
5	$X3 \rightarrow Y$	0,103	0,085	1,447	0,157	H ₇ Not Accepted
6	$X3 \rightarrow Z$	0,374	0,072	6,958	0,014	H ₈ Accepted
7	$Z \rightarrow Y$	0,262	0,067	3,493	0,019	H ₁₀ Accepted
<u>Information:</u> X1: Differentiation Strategy Z: School Image X2: Market Driven M: Culture X3: Artificial intelligence marketing Y: Sustainable Competitive Advantage ^(a) p -value is calculated using the bootstrap bias-corrected percentile method approach						

1. the resulting influence coefficient is 0.238 (positive), meaning that the higher the school's differentiation strategy, the higher the sustainable competitive advantage of the school. Thus, the first hypothesis which states that differentiation strategy has a significant effect on sustainable competitive advantage in private vocational schools in Balikpapan City, can be accepted (H₁ is accepted);
2. the resulting influence coefficient is 0.341 (positive), meaning that the higher the school's differentiation strategy, the stronger the school's image will be. Thus, the second hypothesis which states that differentiation strategy has a significant effect on school image at private vocational schools in Balikpapan City, can also be accepted (H₂ is accepted);
3. the resulting influence coefficient is 0.231 (positive), meaning that the higher the school's market drive, the higher the school's sustainable competitive advantage. Thus,



- the fourth hypothesis which states that market-driven has a significant effect on sustainable competitive advantage in private vocational schools in Balikpapan City, can also be accepted (H4 is accepted);
4. the resulting influence coefficient is 0.311 (positive), meaning that the higher the school's market drive, the stronger the school's image will be. Thus, the fifth hypothesis which states that market-driven has a significant effect on the school image of private vocational schools in Balikpapan City, can also be accepted (H5 is accepted);
 5. the resulting coefficient of influence is only 0.103, meaning that the higher levels of marketing with artificial intelligence have not been able to have a real impact on increasing sustainable competitive advantage in schools. Thus, the seventh hypothesis which states that artificial intelligence marketing has a significant effect on sustainable competitive advantage in private vocational schools in Balikpapan City, cannot be accepted (H7 is rejected);
 6. the resulting influence coefficient is 0.374 (positive), meaning that the better the marketing with artificial intelligence at the school, the stronger the school's image will be. Thus, the eighth hypothesis states that artificial intelligence marketing has a significant effect on school image at private vocational schools in Balikpapan City, and can be accepted (H8 is accepted);
 7. The resulting influence coefficient is 0.262 (positive), meaning that the stronger the school image, the higher the sustainable competitive advantage of the school. Thus, the tenth hypothesis which states that school image has a significant effect on sustainable competitive advantage in private vocational schools in Balikpapan City, can be accepted (H10 is accepted).

Tabel 6

Indirect Effect

No	Indirect Effect	Specific Indirect Effect (Bias-corrected percentile method)			
		Std Estimate	P-value & hipotesis	Types	
1	$X1 \rightarrow Z \rightarrow Y$	0,089	0,015 (Accepted)	(H ₃)	Partially mediation
2	$X2 \rightarrow Z \rightarrow Y$	0,082	0,009 (Accepted)	(H ₆)	Partially mediation
3	$X3 \rightarrow Z \rightarrow Y$	0,098	0,004 (Accepted)	(H ₉)	Fully mediation
Information: X1: Differentiation Strategy X2: Market Driven X3: Artificial intelligence marketing Z: School Image M: Culture Y: Sustainable Competitive Advantage					



1. the results of the indirect path significance test Thus, school image significantly mediates the influence of differentiation strategy on sustainable competitive advantage in private vocational schools in Balikpapan City. The nature of the mediator is partial mediation because the influence can be direct or indirect, this implies that a high differentiation strategy will be able to encourage an increase in sustainable competitive advantage, but if the school's image is also strengthened then the school's sustainable competitive advantage will increase even more;
2. the results of the indirect path significance test Thus, school image also significantly mediates the influence of market-driven on sustainable competitive advantage in private vocational schools in Balikpapan City. The nature of the mediator is partial mediation because the influence can be direct or indirect, this implies that a high market drive will be able to encourage an increase in sustainable competitive advantage, but if the school's image is also strengthened then the school's sustainable competitive advantage will increase even more;
3. the results of the indirect path significance test Thus, school image also significantly mediates the influence of artificial intelligence marketing on sustainable competitive advantage in private vocational schools in Balikpapan City. The nature of the mediator is full mediation because the influence can only be indirect, this implies that marketing with high artificial intelligence must be accompanied by strengthening the school's image so that the school's sustainable competitive advantage can increase.

Figure 3

Total Effect on Sustainable Competitive Advantage

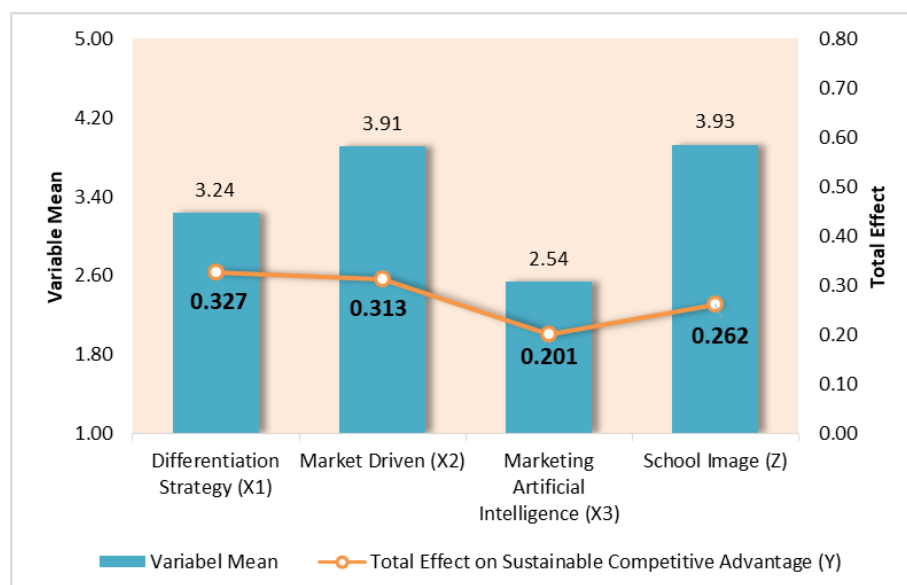




Figure 3. Shows that sustainable competitive advantage in private vocational schools in Balikpapan City is driven more by differentiation strategy (0.327), then market-driven (0.313), school image (0.262), and finally artificial intelligence marketing (0.201). The results of this total effect analysis can provide information regarding the priority scale in efforts to increase sustainable competitive advantage in private vocational schools in Balikpapan City, starting from the highest priority to the lowest priority.

4 DISCUSSION

This research examines the influence of differentiation strategies, market orientation and artificial intelligence-based marketing on sustainable competitive advantage, with school image as a mediating variable, at private vocational schools in Balikpapan City. The research results show that the differentiation strategy has a significant effect on sustainable competitive advantage and school image. This means that the stronger the differentiation strategy implemented by the school, the better the sustainable competitive advantage and school image that is formed. Differentiation strategy is measured through indicators such as product differentiation, human resources, processes, physics, channels, price and place, with place being the most dominant indicator.

Apart from that, market orientation also has a significant influence on sustainable competitive advantage and school image. Market-oriented schools, which prioritize the needs and preferences of students and parents, can increase their competitiveness and positive image. This approach allows schools to remain relevant and responsive to the dynamics of the education market.

Marketing based on artificial intelligence (AI) shows a significant influence on school image but is not significant on sustainable competitive advantage directly. However, through the mediation of school image, AI marketing can contribute to sustainable competitive advantage. The application of AI technology in marketing increases the public's visibility and positive perception of schools, reflecting the school's innovation and progress in utilizing advanced technology.

School image is proven to be an important mediator in the relationship between differentiation strategy, market orientation, and AI marketing towards sustainable competitive advantage. Schools with a strong and positive image can attract more students and maintain customer loyalty, which ultimately strengthens their competitiveness.



School culture, although it does not significantly moderate the influence of school image on sustainable competitive advantage, is still an element that needs to be considered in the long term. A good school culture can strengthen the influence of the school's image on sustainable competitive advantage, even though the impact is not visible in a short time.

Overall, this research confirms the importance of differentiation, market orientation and artificial intelligence-based marketing strategies in building sustainable competitive advantages for private vocational schools in Balikpapan City, with school image as a key factor that mediates this influence. An integrated and responsive approach to market needs and the application of innovative technology will increase the school's competitiveness and reputation sustainably.

5 CONCLUSION

This research concludes that differentiation, market orientation and marketing strategies based on artificial intelligence (AI) play an important role in building sustainable competitive advantage for private vocational schools in Balikpapan City, with school image as a significant mediating variable.

1. **Differentiation Strategy:** This research found that differentiation strategy has a significant influence on sustainable competitive advantage and school image. Schools that implement a differentiation strategy well, through various aspects such as products, human resources, processes, physics, channels, prices and location, can increase their competitiveness sustainably. Location indicators are proven to be the most dominant factor in supporting differentiation;
2. **Market Orientation:** The research results show that market orientation also has a significant effect on sustainable competitive advantage and school image. Market-oriented schools, which understand and respond to the needs and preferences of students and parents, can create a positive image and increase competitiveness;
3. **Artificial Intelligence (AI) Based Marketing:** Although AI-based marketing does not have a direct significant impact on sustainable competitive advantage, its impact on school image has proven to be significant. Through an improved school image, AI-based marketing can contribute indirectly to sustainable competitive advantage;
4. **School Image:** School image plays an important mediating role in the relationship between differentiation strategies, market orientation and AI-based marketing to sustainable competitive advantage. Schools with a strong and positive image can attract



more students and maintain customer loyalty, thereby strengthening their competitiveness;

5. School Culture: Although school culture does not significantly moderate the influence of school image on sustainable competitive advantage, the role of school culture remains important in the long term. A good culture can strengthen the influence of a school's image on sustainable competitive advantage, although the impact may not be visible immediately.

This research confirms that an integrated and responsive approach to market needs, coupled with the application of innovative technologies such as AI, is essential to sustainably improve a school's competitiveness and reputation. Efforts to improve the image of schools, through differentiation strategies and appropriate market orientation, will help private vocational schools in Balikpapan City achieve sustainable competitive advantage.

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